

Active ants: dynamics

$$\begin{cases} \partial_t f = \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) = f_0 \end{cases}$$

- What happens as $t \rightarrow \infty$?

• **Part 1 (Rakotomalala & W, in review, 2025):**

o Existence of compact global attractor

0 Attractor has dimension at least 4 if map is linearly unstable



Linearised around $f = 1/2\pi$, L (not self/skew-adjoint)



Explicit spectral properties for $\mathcal{L}|_{\sigma=0}$

■ Resolvent estimates to $L|_{\sigma>0}$ for small σ



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- What happens as $t \rightarrow \infty$?
- Part 1 (Rakotomalala & dW, in review, 2025):
 - Existence of compact global attractor
 - Attractor has dimensional lower bound of 4 if homogeneous state is linearly unstable
 - Linearised operator around $f = 1/2\pi$, L (**not** self/skew-adjoint)
 - Explicit spectral properties for $L|_{\sigma=0}$
 - Resolvent estimates to translate to $L|_{\sigma>0}$ for small σ



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- What happens as $t \rightarrow \infty$?
- Part 2 (Rakotomalala & dW, in review, 2026):